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Axle assembly having a shift mechanism

Abstract

An axle assembly having a shift mechanism. The shift mechanism includes an actuator that is configured to move the shift collar along the axis to selectively connect a member of a set of drive pinion gears to a drive pinion. A drive member is also provided that is configured to move the shift collar when actuated.

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Background/Summary

TECHNICAL FIELD
(1) This relates to an axle assembly having a shift mechanism for actuating a shift collar.
BACKGROUND
(2) An axle assembly having a clutch collar is disclosed in U.S. Pat. No. 9,719,563.
SUMMARY
(3) In at least one embodiment an axle assembly is provided. The axle assembly includes a drive pinion, a transmission, and a shift mechanism. The drive pinion is rotatable about an axis. The transmission includes a set of drive pinion gears. The drive pinion gears are spaced apart from the drive pinion and are rotatable about the axis. The shift mechanism includes a shift collar, an actuator, a linkage, and a drive member. The shift collar is rotatable about the axis with the drive pinion. The shift collar is moveable along the axis with respect to the drive pinion. The actuator is configured to move the shift collar along the axis to selectively connect a member of the set of drive pinion gears to the drive pinion. The linkage is operatively connected to the actuator and the shift collar. The linkage is rotatable about an actuator axis. The linkage has a linkage gear. The drive member is rotatable about a drive member axis. The drive member has a drive member gear that has teeth that mesh with teeth of the linkage gear. The shift collar is moveable along the axis when the actuator rotates the linkage. The shift collar is moveable along the axis when the drive member is manually actuated to rotate the linkage.
(4) The linkage gear may be a sector gear. The linkage gear may have teeth that extend away from the actuator axis.
(5) The drive member may have a drive member engagement feature. The drive member engagement feature may be engageable with a tool that is configured to manually actuate the drive

member. The drive member gear and the drive member engagement feature may be disposed at opposite ends of the drive member.
(6) The shift mechanism may be received in a shift mechanism housing. The drive member may extend through a hole in the shift mechanism housing. The shift mechanism housing may define a bore. The bore may extend from the hole. A cap may be receivable in the bore to conceal the drive member.
(7) In at least one embodiment an axle assembly is provided. The axle assembly includes a drive pinion, a transmission, and a shift mechanism. The drive pinion is rotatable about an axis. The transmission includes a set of drive pinion gears. The drive pinion gears are spaced apart from the drive pinion. The drive pinion gears are rotatable about the axis. The shift mechanism includes a shift collar, an actuator, a detent linkage, the linkage, and a drive member. The shift collar is rotatable about the axis with the drive pinion. The shift collar is moveable along the axis with respect to the drive pinion. The actuator is configured to move the shift collar along the axis to selectively connect a member of the set of drive pinion gears to the drive pinion. The detent linkage is coupled to the actuator. The detent linkage is rotatable about an actuator axis. The linkage is coupled to the detent linkage. The linkage is operatively connected to the shift collar. The linkage is rotatable about the actuator axis. The drive member is rotatable about the actuator axis with the detent linkage. The shift collar is moveable along the axis when the actuator rotates the linkage. The shift collar is moveable along the axis when the drive member is actuated to rotate the linkage.
(8) The drive member may be received inside the detent linkage. The detent linkage may have a detent linkage spline. The drive member may have a drive member spline. The drive member spline may mate with the detent linkage spline.
(9) The drive member may have a drive member engagement feature. The drive member engagement feature may be configured to be engaged by a tool. The tool may be configured to manually actuate the drive member. The drive member spline and the drive member engagement feature may be disposed at opposite ends of the drive member.
(10) A sensor may be coupled to the drive member. The sensor may generate a signal indicative of a rotational position of the drive member. The sensor may be mounted to the shift mechanism housing. The sensor may be disposed outside the shift mechanism housing.
(11) An adapter may extend from the sensor to the drive member. The adapter may be received inside the drive member. The sensor may have a shaft. The shaft may be received inside the adapter.
(12) The detent linkage may be positioned along the actuator axis closer to the actuator than the linkage is positioned to the actuator. The detent linkage may define a detent linkage hole. The detent linkage hole may receive the linkage. The drive member may define a drive member hole. The drive member hole may receive the linkage. The drive member may have an indicator mark. The indicator mark may provide a visual indication of a position of the shift collar. The visual indication of the position of the shift collar may be visible from outside the shift mechanism housing.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a perspective view of an example of an axle assembly.
(2) FIG. **2** is a section view of the axle assembly along section line **2-2**.
(3) FIG. **3** is an end view of the axle assembly with a portion of cover removed and with an electric motor module and axle housing of the axle assembly omitted for clarity.
(4) FIG. **4** is a section view of a portion of the axle assembly along section line **4-4** with the electric motor module and drive pinion omitted for clarity.

(5) FIG. **5** is a perspective view that includes an example of a shift mechanism having a shift collar that may be provided with the axle assembly.
(6) FIG. **6** is an exploded view of a portion of the shift mechanism shown in FIG. **5**.
(7) FIG. **7** is a side view of a portion of the axle assembly.
(8) FIG. **8** is a section view of a portion of the axle assembly along section line **8-8**.
(9) FIG. **9** is an exploded view of a portion of the shift mechanism shown in FIG. **5**.
(10) FIG. **10** is a section view of a portion of the axle assembly along section line **10-10**.
(11) FIG. **11** is a perspective view of a portion of another configuration of the axle assembly with the electric motor module and drive pinion omitted for clarity.
(12) FIG. **12** is a section view of the portion of the axle assembly along section line **12-12** that includes a second configuration of shift mechanism.
(13) FIG. **13** is a perspective view of the shift mechanism associated with FIGS. **11** and **12**.
(14) FIG. **14** is an exploded view of a portion of the shift mechanism shown in FIG. **13**.
(15) FIG. **15** is a perspective view of a portion of another configuration of the axle assembly with the electric motor module and drive pinion omitted for clarity.
(16) FIG. **16** is a section view of the portion of the axle assembly along section line **16-16** that includes a third configuration of a shift mechanism.
(17) FIG. **17** is a perspective view of the shift mechanism associated with FIGS. **15** and **16**.
(18) FIG. **18** is an exploded view of a portion of the shift mechanism shown in FIG. **17**.
(19) FIG. **19** is a side view of a portion of the axle assembly showing an example of an end of a drive member of the shift mechanism associated with FIGS. **15** and **16**.

DETAILED DESCRIPTION

(20) As required, detailed embodiments of the present invention are disclosed herein; however, it is to be understood that the disclosed embodiments are merely exemplary of the invention that may be embodied in various and alternative forms. The figures are not necessarily to scale; some features may be exaggerated or minimized to show details of particular components. Therefore, specific structural and functional details disclosed herein are not to be interpreted as limiting, but merely as a representative basis for teaching one skilled in the art to variously employ the present invention.
(21) It will also be understood that, although the terms first, second, etc. are, in some instances, used herein to describe various elements, these elements should not be limited by these terms. These terms are only used to distinguish one element from another. For example, a first element could be termed a second element, and similarly a second element could be termed a first element without departing from the scope of the various described embodiments. The first element and the second element are both elements, but they are not the same element.
(22) The terminology used in the description of the various described embodiments is for the purpose of describing particular embodiments only and is not intended to be limiting. As used in the description of the various described embodiments and the appended claims, the singular forms “a”, “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will also be understood that the term “and/or” as used herein refers to and encompasses any and all possible combinations of one or more of the associated listed items. It will be further understood that the terms “includes,” “including,” “comprises,” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.
(23) Referring to FIG. **1**, an example of an axle assembly **10** is shown. The axle assembly **10** may be provided with a motor vehicle like a truck, bus, farm equipment, mining equipment, military transport or weaponry vehicle, or cargo loading equipment for land, air, or marine vessels. The motor vehicle may include a trailer for transporting cargo in one or more embodiments.
(24) The axle assembly **10** is configured to provide torque to one or more traction wheel assemblies

that may include a tire mounted on a wheel. The wheel may be mounted to a wheel hub that may be rotatable about a wheel axis.
(25) One or more axle assemblies may be provided with the vehicle. A single axle assembly is shown in FIGS. **1** and **2**. The axle assembly **10** includes a housing assembly **20**, a differential assembly **22**, at least one axle shaft **24**, an electric motor module **26**, a transmission module **28**, a drive pinion **30**, a shift mechanism **32**, or combinations thereof.
(26) Housing Assembly
(27) Referring to FIG. **1**, the housing assembly **20** receives various components of the axle assembly **10**. In addition, the housing assembly **20** may facilitate mounting of the axle assembly **10** to the vehicle. In at least one configuration, the housing assembly **20** may include an axle housing **40** and a differential carrier **42**.
(28) The axle housing **40** may receive and may support the axle shafts **24**. In at least one configuration, the axle housing **40** may include a center portion **50** and at least one arm portion **52**.
(29) The center portion **50** is disposed proximate the center of the axle housing **40**. As is best shown in FIG. **2**, the center portion **50** may define a cavity **54** that may at least partially receive the differential assembly **22**. A lower region of the center portion **50** may at least partially define a sump portion **56** that may contain or collect lubricant **58**. Lubricant **58** in the sump portion **56** may be splashed by a ring gear **82** of the differential assembly **22** and distributed to lubricate various components that may or may not be received in the housing assembly **20**. For instance, some splashed lubricant **58** may lubricate components that are received in the cavity **54** like the differential assembly **22**, bearing assemblies that rotatably support the differential assembly **22**, a drive pinion **30**, and so on, while some splashed lubricant **58** may be routed out of the cavity **54** to lubricate components located outside of the housing assembly **20**, such as components associated with the transmission module **28**, the shift mechanism **32**, or both.
(30) Referring to FIG. **1**, one or more arm portions **52** may extend from the center portion **50**. For instance, two arm portions **52** may extend in opposite directions from the center portion **50** and away from the differential assembly **22**. The arm portions **52** may have similar configurations. For example, the arm portions **52** may each have a hollow tubular configuration that may extend around and may receive a corresponding axle shaft **24** and may help separate or isolate the axle shaft **24** or a portion thereof from the surrounding environment. An arm portion **52** or a portion thereof may or may not be integrally formed with the center portion **50**. It is also contemplated that the arm portions **52** may be omitted.
(31) Referring primarily to FIG. **2**, the differential carrier **42** is configured to support the differential assembly **22**. For example, the differential carrier **42** may include one or more bearing supports that may support a bearing like a roller bearing assembly that may rotatably support the differential assembly **22**. The differential carrier **42** may be mounted to the center portion **50** of the axle housing **40**. The differential carrier **42** may also facilitate mounting of the electric motor module **26**. In at least one configuration, the differential carrier **42** may include a mounting flange **60** and/or a bearing support wall **62**.
(32) The mounting flange **60** may facilitate mounting of the electric motor module **26**. As an example, the mounting flange **60** may be configured as a ring that may extend around an axis **70**. In at least one configuration, the mounting flange **60** may include a set of fastener holes that may be configured to receive fasteners that may secure the electric motor module **26** to the mounting flange **60**.
(33) The bearing support wall **62** may support bearings that may rotatably support other components of the axle assembly **10**. For example, the bearing support wall **62** may support a bearing that may rotatably support the drive pinion **30**, a bearing that may rotatably support a rotor of the electric motor module **26**, or both. The bearing support wall **62** may extend in an axial direction away from the axle housing **40** and may extend around the axis **70**. The bearing support wall **62** may define a hole that may extend along or around the axis **70** and receive the drive pinion

30 and the bearings that rotatably support the drive pinion **30**. The bearing support wall **62** may be integrally formed with the differential carrier **42** or may be a separate component that is fastened to the differential carrier **42**.

(34) Differential Assembly, Drive Pinion, and Axle Shafts

(35) Referring to FIG. **2**, the differential assembly **22** is rotatable about a differential axis **80** and is configured to transmit torque to the axle shafts **24** and wheels. The differential assembly **22** is operatively connected to the axle shafts **24** and may permit the axle shafts **24** to rotate at different rotational speeds in a manner known by those skilled in the art. The differential assembly **22** may be at least partially received in the center portion **50** of the housing assembly **20**. The differential assembly **22** may have a ring gear **82** that may have teeth that mate or mesh with the teeth of a gear portion of the drive pinion **30**. Accordingly, the differential assembly **22** may receive torque from the drive pinion **30** via the ring gear **82** and transmit torque to the axle shafts **24**.

(36) The drive pinion **30** may operatively connect the transmission module **28** to the differential assembly **22**. As such, the drive pinion **30** may transmit torque between the differential assembly **22** and the transmission module **28**. In at least one configuration, the drive pinion **30** may be rotatable about the axis **70** and may be rotatably supported inside another component, such as the bearing support wall **62**.

(37) Referring primarily to FIGS. **2** and **6**, the drive pinion **30** may optionally include or may be coupled to a drive pinion extension **90**. The drive pinion extension **90** may effectively increase the axial length of the drive pinion **30**. In at least one configuration, the drive pinion extension **90** may be a separate component from the drive pinion **30** and may be coupled to the drive pinion **30** such that the drive pinion extension **90** is rotatable about the axis **70** with the drive pinion **30**. In addition, the drive pinion extension **90** may be fixedly positioned with respect to the drive pinion **30** such that the drive pinion extension **90** may not move along the axis **70** with respect to the drive pinion **30**. It is also contemplated that the drive pinion extension **90** may be integrally formed with the drive pinion **30**. For convenience in reference, the term “drive pinion **30**” is used herein to refer to the drive pinion **30** with or without the drive pinion extension **90**.

(38) In at least one configuration, the drive pinion extension **90** may extend from a first end **92** to a second end **94** and may include a socket **96** and the spline **98**. The socket **96** may extend from the first end **92** and may receive the drive pinion **30**. The second end **94** may be received inside and may be rotatably supported by a support bearing **418**. The spline **98**, if provided, may facilitate coupling of the drive pinion extension **90** to a shift collar **310** that may be moveable along the axis **70** as will be discussed in more detail below.

(39) Referring to FIG. **1**, the axle shafts **24** are configured to transmit torque between the differential assembly **22** and corresponding wheel hubs and wheels. Two axle shafts **24** may be provided such that each axle shaft **24** extends through a different arm portion **52** of axle housing **40**. The axle shafts **24** may extend along and may be rotatable about an axis, such as the differential axis **80**. Each axle shaft **24** may have a first end and a second end. The first end may be operatively connected to the differential assembly **22**. The second end may be disposed opposite the first end and may be operatively connected to a wheel. Optionally, gear reduction may be provided between an axle shaft **24** and a wheel.

(40) Electric Motor Module

(41) Referring to FIG. **2**, the electric motor module **26**, which may also be referred to as an electric motor, is configured to provide propulsion torque. The electric motor module **26** may be mounted to the differential carrier **42** and may be operatively connectable to the differential assembly **22**. For instance, the electric motor module **26** may configured to provide torque to the differential assembly **22** via the transmission module **28** and the drive pinion **30** as will be discussed in more detail below. The electric motor module **26** may be primarily or completely disposed outside the differential carrier **42**. In addition, the electric motor module **26** may be axially positioned between the axle housing **40** and the transmission module **28**. In at least one configuration, the electric

motor module **26** may include a motor housing **100**, a coolant jacket **102**, a stator **104**, a rotor **106**, and at least one rotor bearing assembly **108**. The electric motor module **26** may also include a motor cover **110**.

(42) The motor housing **100** may extend between the differential carrier **42** and the motor cover **110**. The motor housing **100** may be mounted to the differential carrier **42** and the motor cover **110**. For example, the motor housing **100** may extend from the mounting flange **60** of the differential carrier **42** to the motor cover **110**. The motor housing **100** may extend around the axis **70** and may define a motor housing cavity **120**. The motor housing cavity **120** may be disposed inside the motor housing **100** and may have a generally cylindrical configuration. The bearing support wall **62** of the differential carrier **42** may be located inside the motor housing cavity **120**. Moreover, the motor housing **100** may extend continuously around and may be spaced apart from the bearing support wall **62**. In at least one configuration, the motor housing **100** may have an exterior side **122**, an interior side **124**, a first end surface **126**, and a second end surface **128**.

(43) The exterior side **122** faces away from the axis **70** and may define an exterior or outside surface of the motor housing **100**.

(44) The interior side **124** is disposed opposite the exterior side **122** and may face toward the axis **70**. The interior side **124** may be disposed at a substantially constant radial distance from the axis **70** in one or more configurations.

(45) The first end surface **126** is disposed at an end of the motor housing **100** that may face toward the differential carrier **42**. For instance, the first end surface **126** may be disposed adjacent to the mounting flange **60** of the differential carrier **42** and may engage or contact the mounting flange **60**. The first end surface **126** may extend between the exterior side **122** and the interior side **124**.

(46) The second end surface **128** may be disposed opposite the first end surface **126**. As such, the second end surface **128** may be disposed at an end of the motor housing **100** that may face toward the motor cover **110** and may engage or contact the motor cover **110**.

(47) The coolant jacket **102** facilitates cooling or heat removal, such cooling of the stator **104**. The coolant jacket **102** may be received in the motor housing cavity **120** of the motor housing **100** and may engage the interior side **124** of the motor housing **100**. The coolant jacket **102** may extend axially (e.g., in a direction along the axis **70**) between the differential carrier **42** and the motor cover **110**. For example, the coolant jacket **102** may extend axially from the differential carrier **42** to the motor cover **110**. In addition, the coolant jacket **102** may extend around the axis **70** and around the stator **104**. Accordingly, the stator **104** may be at least partially received in and may be encircled by the coolant jacket **102**. The coolant jacket **102** may extend in a radial direction from the stator **104** to the interior side **124** of the motor housing **100**. In at least one configuration, the coolant jacket **102** may include a plurality of channels through which coolant may flow.

(48) The stator **104** is received in the motor housing cavity **120**. The stator **104** may be fixedly positioned with respect to the coolant jacket **102**. For example, the stator **104** may extend around the axis **70** and may include stator windings that may be received inside and may be fixedly positioned with respect to the coolant jacket **102**.

(49) The rotor **106** extends around and is rotatable about an axis, such as axis **70**. In addition, the rotor **106** may extend around and may be supported by the bearing support wall **62**. The rotor **106** may be received inside the stator **104**, the coolant jacket **102**, and the motor housing cavity **120** of the motor housing **100**. The rotor **106** may be rotatable about the axis **70** with respect to the differential carrier **42** and the stator **104**. In addition, the rotor **106** may be spaced apart from the stator **104** but may be disposed in close proximity to the stator **104**.

(50) One or more rotor bearing assemblies **108** rotatably support the rotor **106**. For example, a rotor bearing assembly **108** may extend around and receive the bearing support wall **62** of the differential carrier **42** and may be received inside of the rotor **106**. The rotor **106** may be operatively connected to the drive pinion **30**. For instance, a coupling such as a rotor output flange **130** may operatively connect the rotor **106** to the transmission module **28**, which in turn may be operatively connectable

to the drive pinion **30**.

(51) The motor cover **110** may be mounted to the motor housing **100** and may be disposed opposite the axle housing **40** and the differential carrier **42**. For example, the motor cover **110** may be mounted to the second end surface **128** of the motor housing **100**. The motor cover **110** may be spaced apart from and may not engage the differential carrier **42**. The motor cover **110** may be provided in various configurations. In at least one configuration, the motor cover **110** may include a first side **140** and a second side **142**. The first side **140** may face toward and may engage the motor housing **100**. The second side **142** may be disposed opposite the first side **140**. The second side **142** may face away from the motor housing **100**. The motor cover **110** may also include a motor cover opening through which the drive pinion **30** may extend. The motor cover **110** may be integrated with the transmission module **28** or may be a separate component.

(52) Transmission Module

(53) Referring to FIGS. **2** and **4**, the transmission module **28** is configured to transmit torque between the electric motor module **26** and the differential assembly **22**. As such, the transmission module **28** may be operatively connectable to the electric motor module **26** and the differential assembly **22**. In at least one configuration, the transmission module **28** may include a transmission housing. The transmission housing may include one or more individual housings, such as a first transmission housing **200**, a second transmission housing **202**. The transmission module **28** may also include a transmission **204**. The first transmission housing **200** and the second transmission housing **202** may cooperate to define a transmission housing cavity **206** that may receive the transmission **204**.

(54) The first transmission housing **200** may be mounted to the electric motor module **26**. For instance, the first transmission housing **200** may be mounted to the second side **142** of the motor cover **110**. As such, the motor cover **110** may separate the first transmission housing **200** from the motor housing **100**.

(55) The second transmission housing **202** may be mounted to the first transmission housing **200**. For instance, the first transmission housing **200** may be mounted to and may engage or contact a side of the first transmission housing **200** that may face away from the motor cover **110**. As such, the first transmission housing **200** may separate the second transmission housing **202** from the motor cover **110**.

(56) The transmission **204** may be operatively connected to the electric motor. In at least one configuration, the transmission **204** may be configured as a countershaft transmission that includes a set of drive pinion gears **210**, a first countershaft gear set **212**, and optionally a second countershaft gear set **214**.

(57) The set of drive pinion gears **210** is received in the transmission housing cavity **206** of the transmission housing and may be arranged along the axis **70** between the first transmission housing **200** and the second transmission housing **202**. The set of drive pinion gears **210** may include a plurality of gears, some or all of which may be selectively coupled to the drive pinion **30**. The set of drive pinion gears **210** is spaced apart from the drive pinion **30** and is rotatable about the axis **70**. The gears may be independently rotatable with respect to each other. In the configuration shown, the set of drive pinion gears **210** includes a first gear **220**, a second gear **222**, a third gear **224**, and a fourth gear **226**; however, it is to be understood that a greater or lesser number of gears may be provided.

(58) The first gear **220** extends around the axis **70** and may be disposed proximate the first transmission housing **200**. In at least one configuration, the first gear **220** may have a through hole that may receive the drive pinion **30**, an extension of the drive pinion **30** like the drive pinion extension **90**, or both. The first gear **220** may have a plurality of teeth that may be arranged around and may extend away from the axis **70**. The teeth of the first gear **220** may contact and may mate or mesh with teeth of a first countershaft gear that may be provided with the first countershaft gear set **212** and the second countershaft gear set **214** as will be discussed in more detail below. The first

gear **220** may be operatively connected to the rotor **106** of the electric motor module **26** such that the rotor **106** and the first gear **220** are rotatable together about the axis **70**. For example, the first gear **220** may be fixedly positioned with respect to the rotor **106** or fixedly coupled to the rotor **106** such that the first gear **220** is not rotatable about the axis **70** with respect to the rotor **106**. It is contemplated that the first gear **220** may be fixedly mounted to or integrally formed with the rotor output flange **130**. As such, the first gear **220** may be continuously connected to the rotor **106** such that the first gear **220** and the rotor **106** may be rotatable together about the axis **70** but may not be rotatable with respect to each other. It is also contemplated that the first gear **220** may be selectively coupled to the drive pinion **30** or drive pinion extension **90**, such as with a shift collar. In addition, the first gear **220** may be decoupled from the drive pinion **30** and may be rotatable with respect to the drive pinion **30**. As such, a clutch or shift collar **310** may not connect the first gear **220** to the drive pinion **30** or the drive pinion extension **90**. The drive pinion extension **90**, if provided, may be received inside the first gear **220** and may be spaced apart from the first gear **220**. In at least one configuration, the first gear **220** may be axially positioned along the axis **70** between the second gear **222** and the electric motor module **26**.

(59) The second gear **222** extends around the axis **70**. In at least one configuration, the second gear **222** may have a through hole that may receive the drive pinion **30**, the drive pinion extension **90**, or both. The second gear **222** may have a plurality of teeth that may be arranged around and may extend away from the axis **70**. The teeth of the second gear **222** may contact and may mate or mesh with teeth of a second countershaft gear that may be provided with the first countershaft gear set **212** and the second countershaft gear set **214** as will be discussed in more detail below. As is best shown in FIG. **4**, the second gear **222** may also have inner gear teeth **232** that may extend toward the axis **70** and may be received in the through hole. The second gear **222** may have a different diameter than the first gear **220**. For example, the second gear **222** may have a larger diameter than the first gear **220**. In at least one configuration, the second gear **222** may be axially positioned along the axis **70** between the first gear **220** and the third gear **224**. The drive pinion **30** or drive pinion extension **90**, if provided, may be received inside the second gear **222** and may be spaced apart from the second gear **222** in one or more configurations.

(60) The third gear **224** extends around the axis **70**. In at least one configuration, the third gear **224** may have a through hole that may receive the drive pinion **30**, the drive pinion extension **90**, or both. The third gear **224** may have a plurality of teeth that may be arranged around and may extend away from the axis **70**. The teeth of the third gear **224** may contact and may mate or mesh with teeth of a third countershaft gear that may be provided with the first countershaft gear set **212** and the second countershaft gear set **214** as will be discussed in more detail below. As is best shown in FIG. **4**, the third gear **224** may also have inner gear teeth **234** that may extend toward the axis **70** and may be received in the through hole. The third gear **224** may have a different diameter than the first gear **220** and the second gear **222**. For example, the third gear **224** may have a larger diameter than the first gear **220** and the second gear **222**. In at least one configuration, the third gear **224** be axially positioned along the axis **70** between the second gear **222** and the fourth gear **226**. The drive pinion **30** or drive pinion extension **90**, if provided, may be received inside the third gear **224** and may be spaced apart from the third gear **224** in one or more configurations.

(61) The fourth gear **226** extends around the axis **70**. In at least one configuration, the fourth gear **226** may have a through hole that may receive the drive pinion **30**, the drive pinion extension **90**, or both. The fourth gear **226** may have a plurality of teeth that may be arranged around and may extend away from the axis **70**. The teeth of the fourth gear **226** may contact and may mate or mesh with teeth of a fourth countershaft gear that may be provided with the first countershaft gear set **212** and the second countershaft gear set **214** as will be discussed in more detail below. As is best shown in FIG. **4**, the fourth gear **226** may also have inner gear teeth **236** that may extend toward the axis **70** and may be received in the through hole. The fourth gear **226** may have a different diameter than the first gear **220**, the second gear **222**, and the third gear **224**, such as a larger

diameter. In at least one configuration, the fourth gear **226** be axially positioned along the axis **70** further from the electric motor module **26** than the first gear **220**, the second gear **222**, and the third gear **224**. As such, the fourth gear **226** may be axially positioned proximate or adjacent to a side of the second transmission housing **202** that is disposed opposite the first transmission housing **200**. The drive pinion **30** or drive pinion extension **90** may be received inside the fourth gear **226** and may be spaced apart from the fourth gear **226** in one or more configurations.

(62) Referring to FIG. **4**, thrust bearings **240** may optionally be provided between members of the set of drive pinion gears **210**, between the first transmission housing **200** and the set of drive pinion gears **210**, between the second transmission housing **202** and the set of drive pinion gears **210**, or combinations thereof.

(63) The first countershaft gear set **212** is received in the transmission housing cavity **206** and may be in meshing engagement with the set of drive pinion gears **210**. The first countershaft gear set **212** may be rotatable about a first countershaft axis **250**. The first countershaft axis **250** may be disposed parallel or substantially parallel to the axis **70** in one or more embodiments. The term "substantially parallel" as used herein means the same as or very close to parallel and includes features or axes that are within ±3° of being parallel each other. The first countershaft gear set **212** may include a first countershaft **260** and a plurality of gears. In the configuration shown, the plurality of gears of the first countershaft gear set **212** include a first countershaft gear **270**, a second countershaft gear **272**, a third countershaft gear **274**, and a fourth countershaft gear **276**; however, it is contemplated that a greater number of countershaft gears or a lesser number of countershaft gears may be provided.

(64) The first countershaft **260** is rotatable about the first countershaft axis **250**. For instance, the first countershaft **260** may be rotatably supported on the first transmission housing **200** and the second transmission housing **202** by corresponding bearing assemblies **280**. For example, first and second bearing assemblies **280** may be located near opposing first and second ends the first countershaft **260**, respectively. The first countershaft **260** may support and be rotatable with the first countershaft gear **270**, the second countershaft gear **272**, the third countershaft gear **274**, and the fourth countershaft gear **276**.

(65) The first countershaft gear **270** is fixedly disposed on the first countershaft **260** or fixedly mounted to the first countershaft **260**. As such, the first countershaft gear **270** may rotate about the first countershaft axis **250** with the first countershaft **260** and may not be rotatable with respect to the first countershaft **260**. For example, the first countershaft gear **270** may have a hole that may receive the first countershaft **260** and may be fixedly coupled to the first countershaft **260**. The first countershaft gear **270** may extend around the first countershaft axis **250** and may have a plurality of teeth that may be arranged around and may extend away from the first countershaft axis **250**. The teeth of the first countershaft gear **270** may contact and may mate or mesh with the teeth of the first gear **220**. In at least one configuration, the first countershaft gear **270** may be axially positioned along the first countershaft axis **250** between the first transmission housing **200** and the second countershaft gear **272** of the first countershaft gear set **212**.

(66) The second countershaft gear **272** is fixedly disposed on the first countershaft **260** or fixedly mounted to the first countershaft **260**. As such, the second countershaft gear **272** may rotate about the first countershaft axis **250** with the first countershaft **260** and may not be rotatable with respect to the first countershaft **260**. For example, the second countershaft gear **272** may have a hole that may receive the first countershaft **260** and may be fixedly coupled to the first countershaft **260**. The second countershaft gear **272** may extend around the first countershaft axis **250** and may have a plurality of teeth that may be arranged around and may extend away from the first countershaft axis **250**. The teeth of the second countershaft gear **272** may contact and may mate or mesh with the teeth of the second gear **222**. The second countershaft gear **272** may have a different diameter than the first countershaft gear **270** and the third countershaft gear **274**. In at least one configuration, the second countershaft gear **272** may be axially positioned along the first countershaft axis **250**

between the first countershaft gear **270** of the first countershaft gear set **212** and the third countershaft gear **274** of the first countershaft gear set **212**.

(67) The third countershaft gear **274** is fixedly disposed on the first countershaft **260** or fixedly mounted to the first countershaft **260**. As such, the third countershaft gear **274** may rotate about the first countershaft axis **250** with the first countershaft **260** and may not be rotatable with respect to the first countershaft **260**. For example, the third countershaft gear **274** may have a hole that may receive the first countershaft **260** and may be fixedly coupled to the first countershaft **260**. The third countershaft gear **274** may extend around the first countershaft axis **250** and may have a plurality of teeth that may be arranged around and may extend away from the first countershaft axis **250**. The teeth of the third countershaft gear **274** may contact and may mate or mesh with the teeth of the third gear **224**. The third countershaft gear **274** may have a different diameter than the first countershaft gear **270** and the second countershaft gear **272**. In at least one configuration, the third countershaft gear **274** may be axially positioned along the first countershaft axis **250** between the second countershaft gear **272** of the first countershaft gear set **212** and the fourth countershaft gear **276** of the first countershaft gear set **212**.

(68) The fourth countershaft gear **276** is fixedly disposed on the first countershaft **260** or fixedly mounted to the first countershaft **260**. As such, the fourth countershaft gear **276** may rotate about the first countershaft axis **250** with the first countershaft **260** and may not be rotatable with respect to the first countershaft **260**. For example, the fourth countershaft gear **276** may have a hole that may receive the first countershaft **260** and may be fixedly coupled to the first countershaft **260** or may be integrally formed with the first countershaft **260**. The fourth countershaft gear **276** may extend around the first countershaft axis **250** and may have a plurality of teeth that may be arranged around and may extend away from the first countershaft axis **250**. The teeth of the fourth countershaft gear **276** may contact and may mate or mesh with the teeth of the fourth gear **226**. The fourth countershaft gear **276** may have a different diameter than the first countershaft gear **270**, the second countershaft gear **272**, and the third countershaft gear **274**. In at least one configuration, the fourth countershaft gear **276** may be axially positioned along the first countershaft axis **250** further from the electric motor module **26** than the third countershaft gear **274** of the first countershaft gear set **212**.

(69) The second countershaft gear set **214**, if provided, is received in the transmission housing cavity **206** and may be rotatable about a second countershaft axis **250′**. The second countershaft axis **250′** may be disposed parallel or substantially parallel to the axis **70** and the first countershaft axis **250** in one or more embodiments. The second countershaft gear set **214** may generally be disposed on an opposite side of the axis **70** from the first countershaft gear set **212** or may be disposed such that the first countershaft axis **250** and the second countershaft axis **250′** may be disposed at a common radial distance from the axis **70**. The first and second countershaft gear sets **212**, **214** may be positioned at any suitable rotational angle or position about the axis **70**.

(70) The second countershaft gear set **214** may have the same or substantially the same configuration as the first countershaft gear set **212**. For example, the second countershaft gear set **214** may include a second countershaft **260′** that may be analogous to or may have the same structure as the first countershaft **260**. In addition, the second countershaft gear set **214** may include a plurality of gears that are rotatable with the second countershaft **260′**. In the configuration shown, the plurality of gears of the second countershaft gear set **214** include a first countershaft gear **270′**, a second countershaft gear **272′**, a third countershaft gear **274′**, and a fourth countershaft gear **276′**; however, it is contemplated that a greater number of gears or a lesser number of gears may be provided. The first countershaft gear **270′**, second countershaft gear **272′**, third countershaft gear **274′**, and the fourth countershaft gear **276′** of the second countershaft gear set **214** may be analogous to or may have the same structure as the first countershaft gear **270**, second countershaft gear **272**, third countershaft gear **274**, and the fourth countershaft gear **276**, respectively, of the first countershaft gear set **212**. The first countershaft gear **270′**, second countershaft gear **272′**, third

countershaft gear **274′**, and the fourth countershaft gear **276′** may be arranged along and may be rotatable about a second countershaft axis **250′** rather than the first countershaft axis **250** and may be fixed to the second countershaft **260′** rather than the first countershaft **260**.
(71) The first gear **220** and the first countershaft gears **270**, **270′** may provide a different gear ratio than the second gear **222** and the second countershaft gears **272**, **272′**, the third gear **224** and the third countershaft gears **274**, **274′**, and the fourth gear **226** and the fourth countershaft gears **276**, **276′**. Gear ratios may be provided that are greater than 1:1, less than 1:1, equal (i.e., 1:1), or combinations thereof.
(72) The teeth of the drive pinion gears and the countershaft gears may be of any suitable type. As a non-limiting example, the meshing teeth of the members of the set of drive pinion gears **210**, the gears of the first countershaft gear set **212**, and the gears of the second countershaft gear set **214** may have a helical configuration.
(73) Shift Mechanism
(74) Referring primarily to FIGS. **2**, **5**, and **6**, the shift mechanism **32** is configured to selectively connect a member of the set of drive pinion gears **210** to the drive pinion **30**. For example, the shift mechanism **32** may operatively connect a member of the set of drive pinion gears **210** to the drive pinion **30** to provide torque at a desired gear ratio, and hence may change the torque transmitted between the electric motor module **26** and the differential assembly **22**. The shift mechanism **32** may couple one member of the set of drive pinion gears **210** at a time to the drive pinion **30**. The member of the set of drive pinion gears **210** that is coupled to the drive pinion **30** may be rotatable about the axis **70** with the drive pinion **30**.
(75) The shift mechanism **32** may be received in or partially received in a shift mechanism cavity **300**, which is best shown in FIGS. **2** and **3**. The shift mechanism cavity **300** may be at least partially defined by the second transmission housing **202** may be disposed proximate an end of the axle assembly **10**. Referring to FIGS. **1** and **2**, a cover **302** may enclose an end of the axle assembly **10** and help define the shift mechanism cavity **300**. The cover **302** may be mounted on the end of the second transmission housing **202** to help enclose the shift mechanism cavity **300**. The cover **302** may be a single component or may be an assembly of multiple parts. A portion of the cover **302** is removed in FIG. **3**.
(76) The shift mechanism **32** may have any suitable configuration. In at least one configuration such as is shown in FIG. **5**, the shift mechanism **32** may include a shift collar **310**, an actuator **312**, a detent linkage **314**, a linkage **316**, and a collar **318**. The shift mechanism **32** may also include a biasing member **320**, a first pin **322**, a second pin **324**. In this configuration, the shift mechanism **32** includes a drive member **326**.
(77) Referring primarily to FIGS. **5** and **6**, the shift collar **310** may be rotatable about the axis **70** with the drive pinion **30**. In addition, the shift collar **310** may be moveable along the axis **70** with respect to the drive pinion **30**. The shift collar **310** may selectively connect a member of the set of drive pinion gears **210** to the drive pinion **30** as will be discussed in more detail below. The shift collar **310** may be at least partially received in the shift mechanism cavity **300** and may be extendable through components of the transmission **204**, such as the set of drive pinion gears **210**. In at least one configuration, the shift collar **310** may include a first end **330**, a second end **332**, a shift collar hole **334**, and a shift collar spline **336**. The shift collar **310** may also include a first tubular shift collar portion **340**, a second tubular shift collar portion **342**, a shift collar gear **344**, a threaded portion **348** or combinations thereof.
(78) Referring primarily to FIG. **6**, the first end **330** may face toward the drive pinion **30**. In addition, the first end **330** may be disposed adjacent to the drive pinion **30** or the drive pinion extension **90**.
(79) The second end **332** may be disposed opposite the first end **330**. As such, the second end **332** may face away from the drive pinion **30**.
(80) The shift collar hole **334** may extend along the axis **70** between the first end **330** and the

second end **332**. In at least one configuration, the shift collar hole **334** may be configured as a through hole that may extend from the first end **330** to the second end **332**. The drive pinion **30** or the drive pinion extension **90** may be received inside the shift collar hole **334**.
(81) The shift collar spline **336** may couple the shift collar **310** to the drive pinion **30** or the drive pinion extension **90**. The shift collar spline **336** may be disposed in the shift collar hole **334** and may be axially positioned near the first end **330**. The shift collar spline **336** may extend toward the axis **70** and may mate with a spline of the drive pinion **30** or the spline **98** of the drive pinion extension **90** that may have spline teeth that may extend away from the axis **70**. The mating splines may allow the shift collar **310** to move in an axial direction or along the axis **70** while inhibiting rotation of the shift collar **310** about the axis **70** with respect to the drive pinion **30**. Thus, the shift collar **310** may be rotatable about the axis **70** with the drive pinion **30** when the shift collar spline **336** mates with the spline of the drive pinion **30** or the drive pinion extension **90**.
(82) The first tubular shift collar portion **340** may extend from the first end **330** toward the second end **332**. The first tubular shift collar portion **340** may have a hollow tubular configuration and may be at least partially received inside the set of drive pinion gears **210** of the transmission **204**. The first tubular shift collar portion **340** may have a larger outside diameter than the second tubular shift collar portion **342**.
(83) The second tubular shift collar portion **342**, if provided, may extend from the second end **332** toward the first tubular shift collar portion **340** or to the first tubular shift collar portion **340**. For instance, the second tubular shift collar portion **342** may have a hollow tubular configuration and may be at least partially disposed outside of the set of drive pinion gears **210**.
(84) The shift collar gear **344** may be disposed between the first end **330** and the second end **332** of the shift collar **310**. In at least one configuration, the shift collar gear **344** may be disposed opposite the shift collar hole **334** and may extend from the first tubular shift collar portion **340**. The shift collar gear **344** may have teeth that may be arranged around the axis **70** and that may extend away from the axis **70** and away from the shift collar hole **334**. The shift collar spline **336** may be disposed opposite the shift collar gear **344**. The shift collar gear **344** is engageable with different members of the set of drive pinion gears **210** as will be discussed in more detail below.
(85) The threaded portion **348** may be axially positioned between the first end **330** and the second end **332**. For instance, the threaded portion **348** may be provided with the second tubular shift collar portion **342** and may be axially positioned between the first tubular shift collar portion **340** and the second end **332**. The threaded portion **348** may be disposed on an exterior side of the second tubular shift collar portion **342** that may face away from the axis **70**. It is also contemplated that the threaded portion **348** may be omitted.
(86) Referring to FIG. **5**, the actuator **312** is configured to move the shift collar **310** along the axis **70** to selectively connect a member of the set of drive pinion gears **210** to the drive pinion **30**. The actuator **312** may be of any suitable type, such as an electrical, electromechanical, or mechanical actuator. In at least one configuration, the actuator **312** may be mounted to the second transmission housing **202**. A portion of the actuator **312** may be rotatable about an actuator axis **350**. For instance, the actuator **312** may have an actuator shaft **352** that may extend along the actuator axis **350** and may be rotatable about the actuator axis **350**. The actuator shaft **352** may be operatively connected to the detent linkage **314**.
(87) The detent linkage **314** is coupled to the actuator **312**. For instance, the detent linkage **314** may be fixedly coupled to the actuator shaft **352**. As such, the detent linkage **314** may be rotatable about the actuator axis **350** with the actuator shaft **352**. The detent linkage **314** may define a plurality of recesses **360**. The recesses **360** may be configured to receive a detent feature **362**. The detent feature **362** may inhibit rotation of the detent linkage **314** about the actuator axis **350** when the detent feature **362** is received in a recess **360**. For example, rotation of the detent linkage **314** may be inhibited when the detent feature **362** is in a recess **360** and a sufficient actuation force is not provided by the actuator **312** to overcome the rotational resistance exerted by the detent feature

362. In at least one configuration, the linkage **316** be rotatably disposed on the detent linkage **314** and rotatable about the actuator axis **350**. The linkage **316** may normally rotate with the detent linkage **314** but may rotate with respect to the detent linkage **314** when a blocked shift condition is present.

(88) The linkage **316** may operatively connect the actuator **312** to the shift collar **310**. In at least one configuration, the linkage **316** may be positioned along the actuator axis **350** closer to the actuator **312** than the detent linkage **314** is positioned to the actuator **312**. In this configuration, the linkage **316** may be rotatable about the actuator axis **350** and may be rotatable with respect to the detent linkage **314**. The linkage **316** may have a linkage gear **370**.

(89) Referring primarily to FIG. **9**, the linkage gear **370** may include a plurality of teeth that extend away from the actuator axis **350**. In the configuration shown, the linkage gear **370** is configured as a sector gear that extends along an arc or angular distance of approximately 90°. In at least one configuration, the linkage gear **370** may be disposed on a side of the linkage **316** that faces toward the actuator **312**.

(90) Referring to FIGS. **5** and **6**, the collar **318** may receive the shift collar **310**. The collar **318** may extend at least partially around the axis **70** in the shift collar **310**. For instance, the collar **318** may be configured as a ring that may extend around the axis **70**. The collar **318** may be coupled to the linkage **316** as will be discussed in more detail below. In at least one configuration and as is best shown in FIG. **6**, the collar **318** may include a first collar side **440**, a second collar side **442**, and a collar hole **444**. The collar **318** may also include a collar arm **446** and a shift block **448**.

(91) The first collar side **440** may face toward the transmission module **28**, the drive pinion **30**, or both.

(92) The second collar side **442** may be disposed opposite the first collar side **440**. As such, the second collar side **442** may face away from the transmission module **28**, the drive pinion **30**, or both.

(93) The collar hole **444** may extend between the first collar side **440** and the second collar side **442**. The collar hole **444** may be a through hole that may extend through the collar **318**. The shift collar **310** is received inside the collar hole **444** and may be rotatable about the axis **70** with respect to the collar **318**. For instance, the second tubular shift collar portion **342** may be received inside the collar hole **444** and may extend through the collar hole **444**. In at least one configuration, the collar hole **444** may receive a bearing assembly that may be positioned between the shift collar **310** and the collar **318**. For example, the bearing assembly may extend from an outside circumference of the second tubular shift collar portion **342** to the inside diameter of the collar **318** that defines the collar hole **444**.

(94) The collar arm **446** extends from the collar **318**. For instance, the collar arm **446** may extend from the collar **318** in a direction that extends away from the axis **70**. In the configuration shown, the collar arm **446** is shown extending at an oblique angle from the collar **318** and is angled away from the transmission **204**; however, it is contemplated that the collar arm **446** may be angled toward the transmission **204** or may be disposed substantially perpendicular to the axis **70**. The term “substantially perpendicular” is used herein to designate features or axes that are the same as or very close to perpendicular and includes features that are within ±3° of being perpendicular each other. The collar arm **446** may be integrally formed with the collar **318** or may be a separate component that is fastened to the collar **318**. In the configuration shown, the collar arm **446** is illustrated as being integrally formed with the collar **318** and is disposed below the axis **70**. The collar arm **446** is moveably disposed on an alignment rod **450**. The collar arm **446** and the alignment rod **450** may cooperate to limit or inhibit rotation of the collar **318** about the axis **70**.

(95) The alignment rod **450** is disposed on the shift mechanism cavity **300**. For instance, the alignment rod **450** may be received in the shift mechanism cavity **300** and may be mounted to the second transmission housing **202**, the cover **302**, or both. In the configuration shown, the alignment rod **450** is shown as being received in a pocket or recess in the cover **302** and in a pocket of the

second transmission housing **202**. The alignment rod **450** may be fixedly disposed on the cover **302** or the second transmission housing **202** or may be disposed in a manner in which movement of the alignment rod **450** is limited. For example, the alignment rod **450** may slide along the axis **70** and/or rotate about the axis **70** but may remain its axial orientation. The alignment rod **450** may be disposed substantially parallel to the axis **70**. In at least one configuration, the alignment rod **450** may be disposed below the axis **70**, below the shift collar **310**, or both. In at least one configuration, the collar arm **446** has an opening in which the alignment rod **450** may be received. The opening may be a hole, recess, slot or the like inside which the alignment rod **450** may be received. It is also contemplated that the alignment rod **450** may define a recess or slot that extends along its axial length and a portion of the alignment rod **450**, such as the end of the alignment rod **450**, may be received in the recess or slot in the alignment rod **450**.

(96) Referring to FIG. **6**, the shift block **448**, if provided, may be fixedly positioned with respect to the collar **318**. The shift block **448** may be integrally formed with the collar **318** or may be provided as a separate component that is attached to the collar **318**. For instance, the shift block **448** may extend from an outside circumference of the collar **318**, the second collar side **442**, or combinations thereof. The shift block **448**, if provided, may facilitate mounting of a fastener **452** that may connect or couple the linkage **316** to the collar **318**.

(97) The first thrust bearing **410** may facilitate rotation of the shift collar **310** about the axis **70** with respect to the collar **318**. The first thrust bearing **410** may be axially positioned between the first collar side **440** and the shift collar **310**. Optionally, washers may be axially positioned adjacent to one or both sides of the first thrust bearing **410**.

(98) The second thrust bearing **412** may facilitate rotation of the shift collar **310** about the axis **70** with respect to the collar **318**. The second thrust bearing **412** may be positioned between the second collar side **442** and the retainer nut **414**. Optionally a washer may be axially positioned adjacent to one or both sides of the second thrust bearing **412**. For example, a washer may be provided between the second thrust bearing **412** and the retainer nut **414**.

(99) The retainer nut **414** may be mounted to the shift collar **310**. For instance, the retainer nut **414** may have a threaded hole that may receive the second tubular shift collar portion **342** and mate with the threaded portion **348** of the shift collar **310**. The retainer nut **414** may inhibit axial movement of the shift collar **310** with respect to the collar **318** and may help secure the first thrust bearing **410** and the second thrust bearing **412**. It is also contemplated that the retainer nut **414** may be omitted and a different fastener or fastening technique may be used. For instance, a fastener like a snap ring or a press-fit fastener may replace a threaded connection.

(100) An encoder disc **416** may optionally be mounted to the drive pinion **30** or the drive pinion extension **90**. In at least one configuration, the encoder disc **416** may be disposed adjacent to the retainer nut **414**. For instance, the encoder disc **416** may be axially positioned between the retainer nut **414** and a support bearing **418** that rotatably supports the drive pinion **30** or drive pinion extension **90**. For example, the encoder disc **416** may be positioned between a shoulder of the drive pinion **30** or drive pinion extension **90** and the support bearing **418**, if provided. The encoder disc **416** may have detectable features such as protrusions and/or recesses that may be detectable by a sensor to detect rotation or the rotational speed of the drive pinion **30**.

(101) The support bearing **418** may rotatably support the drive pinion **30** or drive pinion extension **90**. For instance, the drive pinion **30** or drive pinion extension **90** may be received inside and may be rotatably supported by the support bearing **418**, which in turn may be supported by the second transmission housing **202**, the cover **302**, or both.

(102) Referring to FIG. **5**, the biasing member **320** may operatively connect the detent linkage **314** to the linkage **316**. In addition, the biasing member **320** may control relative rotational movement between the detent linkage **314** and the linkage **316** (e.g., rotational movement of the linkage **316** with respect to the detent linkage **314**). For example, the biasing member **320** may permit the actuator shaft **352** and the detent linkage **314** to rotate about the actuator axis **350** with respect to

the linkage **316** when the shift collar **310** is inhibited from moving along the axis **70**, such as during a blocked shift as will be discussed in more detail below. The biasing member **320** may be positioned along the actuator axis **350** between the detent linkage **314** and the linkage **316**. The biasing member **320** may have any suitable configuration. For instance, the biasing member **320** may be configured as a spring, such as a torsion spring.

(103) The first pin **322** may extend from the detent linkage **314** toward the linkage **316**. The first pin **322** may be spaced apart from the linkage **316**. The first pin **322** may engage an end or tab of the biasing member **320**.

(104) The second pin **324** may extend from the linkage **316**. The second pin **324** may be spaced apart from the first pin **322** and the detent linkage **314**. The second pin **324** may engage the same end or tab and/or a different end or tab of the biasing member **320**.

(105) Referring to FIGS. **5**, **7**, **9**, and **10**, the drive member **326** is rotatable about a drive member axis **460**. For example, as is best shown in FIG. **10** the drive member **326** may be rotatably disposed on a bearing or bushing **462** that may be received in a hole **464** in a housing, such as a shift mechanism housing **466** that at least partially defines the shift mechanism cavity **300**. The shift mechanism housing **466** may be the second transmission housing **202**, a portion of the second transmission housing **202**, or a housing component that is separate from the second transmission housing **202** and that may be mounted to the second transmission housing **202**. Opposing ends of the bushing **462** may engage corresponding washers **468**. The washers **468** may be disposed outside of the hole **464**.

(106) A retainer **470**, such as a snap ring, clip, fastener, or the like, may be mounted on the drive member **326** and may engage a washer **468** to inhibit axial movement of the drive member **326**. The drive member **326** may extend through the hole **464** such that opposing ends of the drive member **326** are disposed outside of the hole **464**.

(107) Referring to FIGS. **9** and **10**, the drive member **326** may include a drive member gear **480** and a drive member engagement feature **482**.

(108) The drive member gear **480** has teeth that are arranged around the drive member axis **460**. The teeth may extend away from the drive member axis **460** and that mesh with teeth of the linkage gear **370**. The drive member gear **480** may be disposed at an end of the drive member **326** and may be received inside the shift mechanism cavity **300**.

(109) The drive member engagement feature **482** may be disposed at an end of the drive member **326**. For instance, the drive member engagement feature **482** may be disposed at an opposite end of the drive member **326** from the drive member gear **480**. The drive member engagement feature **482** may have any suitable configuration. For instance, the drive member engagement feature **482** may have a male configuration, female configuration, or combinations thereof. In the configuration shown in FIG. **10**, the drive member engagement feature **482** has a female configuration. The drive member engagement feature **482** may be engageable with a tool **484**, which is shown in phantom in FIG. **10**. The tool **484** may be used to manually actuate the drive member **326**. The tool **484** may have any suitable configuration. For instance, the tool **484** may be a driver bit, socket, screwdriver, or the like. The term manual actuation includes both force that is exerted by a person and force that is exerted by device, such as a power tool that may be electrically or pneumatically powered or other actuator that may not be mounted on the axle assembly **10**.

(110) A cap **490** may be removably mounted on the shift mechanism housing **466**. The cap **490** may cover or conceal the drive member **326** when the cap **490** is mounted to the shift mechanism housing **466**. The cap **490** may be removed to permit access to the drive member **326**. The cap **490** may be received in a bore **492** that extends from the hole **464** in the shift mechanism housing **466**. The bore **492** may have a larger diameter than the hole **464**. The cap **490** may be spaced apart from the drive member **326**, bushing **462**, washers **468**, retainer **470**, bore **492**, or combinations thereof. The cap **490** may have any suitable configuration. For instance, the cap **490** may be externally threaded and may mate with a corresponding threaded wall that defines the bore **492**. It is also

contemplated that the cap **490** may not be threaded and may be secured in the bore **492** with an interference fit or a retainer **494**, such as a snap ring, clip, fastener, or the like. In the configuration shown in FIG. **10**, a retainer **494** is received in a groove that extends from the bore **492**. The cap **490** may also include a handle **496** that may be grasped to facilitate insertion and removal of the cap **490**.

(111) The drive member **326** may be manually actuated to rotate the linkage **316**. For instance, the cap **490**, if provided, may be removed from the bore **492**. Next, the tool **484** may be aligned with the drive member engagement feature **482** and engaged with the drive member engagement feature **482**, such as by moving the tool **484** along the drive member axis **460** toward the drive member **326** and into the bore **492**. Next, the tool **484** may be rotated about the drive member axis **460**, such as by manually rotating the tool **484** or by rotating the tool with a power tool or other actuator as previously discussed. The tool **484** may be rotated in a clockwise direction or counterclockwise direction about the drive member axis **460** depending on the direction in which one wants to move the shift collar **310** along the axis **70**. The shift collar **310** is moveable along the axis **70** when the drive member **326** is rotated. More specifically, rotating the drive member **326** causes the linkage **316** to rotate about the actuator axis **350** due to the meshing of the drive member gear **480** and the linkage gear **370**. Rotation of the linkage **316** in turn causes the shift collar **310** to move along the axis **70**.

(112) It is contemplated that manual actuation of the drive member **326** may be employed for maintenance purposes or during assembly to help position the shift collar **310** along the axis **70** independently from operation of the actuator **312**. As such, the shift collar **310** is moveable along the axis **70** when the actuator **312** rotates, in which case the drive member **326** may not be manually actuated, and the shift collar **310** is also moveable along the axis **70** when the drive member **326** is manually actuated, in which case the actuator **312** may not be operated to actuate the shift collar **310**.

(113) Various components of the shift mechanism **32** may typically move together when the shift collar **310** is free to move along the axis **70**. For instance, components such as the actuator shaft **352**, detent linkage **314**, linkage **316**, and the biasing member **320** may rotate together about the actuator axis **350** when the actuator shaft **352** is rotated and the shift collar **310** is free to move along the axis **70**. However, some of these components may move respect to each other when the shift collar **310** is not free to move along the axis **70**. For example, the actuator shaft **352** and the detent linkage **314** may be rotatable with respect to the linkage **316** when the shift collar **310** is not free to move along the axis **70**. The shift collar **310** may not be free to move along the axis **70** when the rotational speed of the shift collar **310** about the axis **70** is not sufficiently synchronized with the rotational speed of a member of the set of drive pinion gears **210**. For instance, the shift collar **310** may be blocked from shifting or moving along the axis **70** when the teeth of the shift collar gear **344** are inhibited from entering the gaps between the inner gear teeth of a drive pinion gear or exiting the gaps between the inner gear teeth of a drive pinion gear.

(114) Relative rotational movement of the detent linkage **314** with respect to the linkage **316** is accommodated by the biasing member **320** when there is a blocked shift. For instance, the first pin **322** may remain in engagement with the second end or second tab of the biasing member **320** but may be rotated to disengage or move away from the first end or first tab of the biasing member **320**. The second pin **324** may remain in engagement with the first end but may be disengaged from the second end. This relative rotational movement may store potential energy in the biasing member **320**. The potential energy may be released when the blocked shift condition is no longer present, such as when the rotational speed of the shift collar **310** is sufficiently synchronized with the rotational speed of a member of the set of drive pinion gears **210** to permit axial movement of the shift collar **310**. As a result, the actuator **312** may complete its intended rotation of the actuator shaft **352** as if the shift collar **310** not blocked even when a blocked shift condition is present, thereby avoiding heating/overheating of the actuator **312** and the consumption of energy that would

occur if the actuator **312** had to continuously work or exert force to attempt to complete shifting of the shift collar **310**. Moreover, sufficient potential energy may be stored in the biasing member **320** that may be released to complete a shift of the shift collar **310** when sufficient synchronization is obtained.
(115) Referring to FIGS. **11-14**, a portion of another configuration of an axle assembly is shown. This configuration employs the same housing assembly **20**, a differential assembly **22**, axle shafts **24**, the electric motor module **26**, transmission module **28**, and drive pinion **30**, and related components as previously described but has a different configuration for some components of the shift mechanism and the shift mechanism housing. For clarity, the shift mechanism associated with this configuration is designated with reference number **32**′ and the shift mechanism housing is designated with reference number **466**′.
(116) Referring to FIG. **13**, the shift mechanism **32**′ employs the same shift collar **310**, actuator **312**, collar **318**, first pin **322**, and second pin **324** as previously described. This configuration also includes detent linkage **314**′, a linkage **316**′, and a drive member **326**′ and may include a connecting block **320**′. The biasing member **320** is not included in this configuration.
(117) The detent linkage **314**′ is the same as the detent linkage **314** previously described except that the detent linkage **314**′ may be configured to be coupled to the drive member **326**′. The detent linkage **314**′ may be coupled to the drive member **326**′ in any suitable manner. For instance, the detent linkage **314**′ may have a male configuration while the drive member **326**′ may have a female configuration or vice versa. In the configuration shown, the drive member **326**′ is received inside the detent linkage **314**′. For example, the detent linkage **314**′ may have a detent linkage spline **500**, which is best shown in FIGS. **12** and **14**, that may mate with a corresponding drive member spline **510** on the drive member **326**′. The detent linkage spline **500** may be arranged around the actuator axis **350** and may be disposed in a hole in the detent linkage **314**′.
(118) The linkage **316**′ is the same as the linkage **316** previously described except for two main differences. First, the linkage **316**′ does not have a linkage gear **370**. Second, the linkage **316**′ is not rotatable about the actuator axis **350** with respect to the detent linkage **314**′. Instead, the detent linkage **314**′ rotates with the linkage **316**′. The detent linkage **314**′ may be fixedly connected to the linkage **316**′ in any suitable manner. For instance, the detent linkage **314**′ and the linkage **316**′ may be integrally formed, welded, fastened together such as with one or more fasteners, or the like. In the configuration shown, a connecting block **320**′ facilitates coupling of the detent linkage **314**′ and the linkage **316**′. For example, the connecting block **320**′ may be disposed between the detent linkage **314**′ and the linkage **316**′ and may receive the first pin **322** and the second pin **324** to inhibit rotation of the detent linkage **314**′ with respect to linkage **316**′.
(119) The drive member **326**′ is rotatable about the actuator axis **350** with the detent linkage **314**′. In at least one configuration, the drive member **326**′ may include a drive member spline **510**, a drive member engagement feature **512**, a first hub **514**, a second hub **516**, or combinations thereof.
(120) The drive member spline **510** may be configured to mate with the detent linkage spline **500**. In at least one configuration, the drive member spline **510** may extend between a first end of the drive member **326**′ and the first hub **514**. In such a configuration, the drive member spline **510** may be received inside the hole in the detent linkage **314**′. It is also contemplated that the drive member spline **510** may have a female configuration that encircles a corresponding spline on the detent linkage **314**′ rather than a male configuration as shown.
(121) The drive member engagement feature **512** may be analogous to the drive member engagement feature **482** previously discussed. For instance, the drive member engagement feature **512** may be disposed at a second end of the drive member **326**′ that may be disposed opposite the first end. As such, the drive member engagement feature **512** and the drive member spline **510** may be disposed at opposite ends of the drive member **326**′. The drive member engagement feature **512** may have any suitable configuration. For instance, the drive member engagement feature **512** may have a male configuration, female configuration, or combinations thereof. In the configuration

shown in FIG. **14**, the drive member engagement feature **512** has a female configuration. The drive member engagement feature **512** may be engageable with a tool **484** that may be used to manually actuate the drive member **326′**.
(122) The first hub **514** may be configured as a protrusion or annular ring that may extend away from the actuator axis **350**. The first hub **514** may limit axial movement of the drive member **326′** in a direction that extends toward the actuator **312**. For instance, the first hub **514** may have a larger diameter than the hole in the detent linkage **314′** and may engage a side of the detent linkage **314′** that faces away from the actuator **312** to inhibit axial movement of the drive member **326′** toward the actuator **312**.
(123) The second hub **516** may be spaced apart from the first hub **514**. The second hub **516** may be configured as a protrusion or annular ring that may extend away from the actuator axis **350**. The second hub **516** may limit axial movement of the drive member **326′** in a direction that extends away from the actuator **312**. For instance, the second hub **516** may have a larger diameter than a component that receives or encircles the drive member **326′**, such as a bushing or bearing **520** that may rotatably support the drive member **326′**, and may engage a side of such a component that faces toward the actuator **312** to inhibit axial movement of the drive member **326′** away from the actuator **312**.
(124) Referring to FIG. **12**, the bearing **520** may be received in a hole **464′** in the shift mechanism housing **466′**. A retainer **522**, such as a snap ring clip or the like, may engage an outer race of the bearing **520** to secure the bearing **520** in the hole **464′**. For instance, a retainer **522** may be disposed on a side of the bearing **520** that is disposed opposite the second hub **516** and may inhibit or limit axial movement of the bearing **520** away from the actuator **312**. In addition, a portion of the drive member **326′** that is encircled by the bearing **520** and optionally the second hub **516** may be received in the hole in the shift mechanism housing **466′**. Thus, a portion of the drive member **326′** may be received in or extend through the hole **464′** in the shift mechanism housing **466′**. The shift mechanism housing **466′** may be similar or the same as the shift mechanism housing **466** previously discussed except for the location of the hole that receives the drive member **326**.
(125) Also as previously discussed, the shift mechanism housing **466′** may define a bore **492′** that extends from the hole **464′** in the shift mechanism housing **466′**. The bore **492′** may have a larger diameter than the hole **464′**. The bore **492′** may receive a cap **490** and optionally a retainer **494** as previously described. The hole **464′** and the bore **492′** may be the same as or substantially similar to the hole **464** and the bore **492** previously described but may be disposed along the actuator axis **350** rather than along a drive member axis that is disposed parallel to and is offset from the actuator axis **350**.
(126) The drive member **326′** may be manually actuated to rotate a shaft of the actuator **312**, detent linkage **314′** and the linkage **316′** and thus to actuate the shift collar **310**. Rotation of the linkage **316′** cause the shift collar **310** to move along the axis **70**. It is contemplated that manual actuation of the drive member **326′** may be employed for maintenance purposes or during assembly to help position the shift collar **310** along the axis **70**. Steps associated with manually actuating the drive member **326′** with a tool **484** may be similar or the same as the steps previously discussed.
(127) Referring to FIGS. **15-19**, a portion of another configuration of an axle assembly is shown. This configuration is similar to the configuration shown in FIGS. **11-14** in which the drive member is disposed along the actuator axis **350**. This configuration employs the same housing assembly **20**, a differential assembly **22**, axle shafts **24**, electric motor module **26**, transmission module **28**, and drive pinion **30**, and related components as previously described but has a different configuration for some components of the shift mechanism and the shift mechanism housing. For clarity, the shift mechanism associated with this configuration is designated with reference number **32″** and the shift mechanism housing is designated with reference number **466″**.
(128) Referring to FIG. **17**, the shift mechanism **32″** employs the same shift collar **310**, actuator **312**, collar **318**, first pin **322**, and second pin **324** as previously described. This configuration also

includes detent linkage **314**″, a linkage **316**″, a biasing member **320**″, and a drive member **326**″. In addition, this configuration may also include a sensor **550** and optionally an adapter **552**.
(129) Referring primarily to FIGS. **17** and **18**, the detent linkage **314**″ is coupled to the actuator **312** and is rotatable about the actuator axis **350** with the actuator shaft **352**. In addition, the detent linkage **314**″ may define a plurality of recesses **360** as previously described. In this configuration, the detent linkage **314**″ is axially positioned between the actuator **312** and the linkage **316**″. Thus, the detent linkage **314**″ may be positioned along the actuator axis **350** closer to the actuator **312** than the linkage **316**″ is positioned to the actuator **312**. The detent linkage **314**″ may be coupled to the actuator **312** and the linkage **316**″ with a male configuration, female configuration, or combinations thereof. In the configuration shown, the detent linkage **314**″ receives the actuator shaft **352** and the linkage **316**″. For instance, the detent linkage **314**″ may include a detent linkage hole **560** that receives the actuator shaft **352** and the linkage **316**″ or separate detent linkage holes that receive the actuator shaft **352** and the linkage **316**″, respectively. In at least one configuration, the detent linkage **314**″ may have splines that mate with corresponding splines on the actuator shaft **352** and may have a smooth spline-free surface that engages the linkage **316**″.
(130) The linkage **316**″ may operatively connect the actuator **312** to the shift collar **310**. In this configuration, the linkage **316**″ may be rotatable about the actuator axis **350** and may be rotatable with respect to the detent linkage **314**″. In the configuration shown, the linkage **316**″ is axially positioned between and engages the detent linkage **314**″ and the drive member **326**″. The linkage **316**″ may have a nonplanar or bent configuration as compared to the linkages **316**, **316**′ previously described. For instance, the linkage **316**″ may extend downward toward the axis **70** and may bend toward the actuator **312** and extend away from the axis **70**. Then the linkage **316**″ may bend again in a direction that extends away from the actuator axis **350** and thereby position the end of the linkage **316**″ in the same position as the linkages previously described. The linkage **316**″ may interface with the detent linkage **314**″ and the drive member **326**″ with a male configuration, female configuration, or combinations thereof. In the configuration shown, the linkage **316**″ has a male configuration that is received inside the detent linkage **314**″ and the drive member **326**″. The linkage **316**″ may be connected to the drive member **326**″ such that the linkage **316**″ and the drive member **326**″ are rotatable together about the actuator axis **350**.
(131) The biasing member **320**″ may be similar to or the same as the biasing member **320** previously described. The biasing member **320**″ may operatively connect the detent linkage **314**″ to the linkage **316**″.
(132) The biasing member **320**″ may permit the actuator shaft **352** and the detent linkage **314**″ to rotate about the actuator axis **350** with respect to the linkage **316**″ when the shift collar **310** is inhibited from moving along the axis **70**, such as during a blocked shift as previously discussed. The biasing member **320**″ may have first and second ends or tabs that may engage the first pin **322** and the second pin **324**, respectively, as previously discussed.
(133) The drive member **326**″ is rotatable about the actuator axis **350** with the detent linkage **314**″. Referring primarily to FIGS. **16** and **18**, the drive member **326**″ may include a drive member spline **510**″ and a drive member flange **562**″.
(134) The drive member spline **510**″ may mate with a corresponding spline on the linkage **316**″. In the configuration shown, the drive member spline **510**″ is disposed inside a hole or bore in the drive member **326**″.
(135) The drive member flange **562**″ may be disposed at or near an end of the drive member **326**″ that may be disposed opposite the detent linkage **314**″. For instance, the drive member flange **562**″ may be disposed in the bore **492**″ in the shift mechanism housing **466**″. The drive member flange **562**″ may extend away from the actuator axis **350** and may be larger than the hole **464**″ in the shift mechanism housing **466**″ through which the drive member **326**″ extends, thereby inhibiting axial movement of the drive member **326**″ toward the actuator **312**. Alternatively or in addition, the drive member flange **562**″ may be larger than a hole in a bushing that is disposed in the hole **464**″ in the

shift mechanism housing **466″**.

(136) The sensor **550** is configured to generate a signal indicative of a rotational position of the drive member **326″**. The sensor **550** may effectively replace the cap associated with the previous configurations. The sensor **550** may be coupled to the drive member **326″**. For instance, the sensor **550** may have a sensor housing **570** that may be mounted to or secured to the shift mechanism housing **466″**. In the configuration shown, the sensor **550** and more specifically the sensor housing **570** is disposed outside of the shift mechanism housing **466″**. The sensor **550** may have a sensor shaft **572** that may be connectable to the drive member **326″**. For instance, the sensor shaft **572** may extend from the sensor housing **570** and that may be rotatable about the actuator axis **350** with respect to the sensor housing **570**. The sensor shaft **572** may be directly or indirectly coupled to the drive member **326″**. In the configuration shown, the sensor **550** and the sensor shaft **572** are indirectly coupled to the drive member **326″** with the adapter **552**.

(137) The adapter **552**, if provided, may extend from the sensor **550** to the drive member **326″**. The adapter **552** may interface with the sensor shaft **572** in any suitable manner, such as with a male configuration, female configuration, or combinations thereof. In the configuration shown, the adapter **552** is received inside the drive member **326″** and the sensor shaft **572** is received inside a hole in the adapter **552**. As such, the sensor shaft **572**, the adapter **552**, and the drive member **326″** may be rotatable together about the actuator axis **350**.

(138) Referring to FIG. **19**, a magnified side view is shown with the sensor **550** removed to reveal the drive member flange **562″**. The drive member flange **562″** may be provided with an indicator mark **580**. The indicator mark **580** may provide a visual indication of the rotational position of the drive member **326″** and hence may indicate the position of the shift collar **310** in a manner that is visible from outside the shift mechanism housing **466″**. For example, the shift mechanism housing **466″** may include one or more markings **582** that may represent an axial position of the shift collar **310**. In the configuration shown, the numbers **1**, **2**, and **3** represent positions in which the shift collar **310** are engaged with three different drive pinion gears. The letter “N” may represent a neutral position in which the shift collar **310** is not engaged with any of the drive pinion gears. In the example shown, the indicator mark **580** is aligned with a neutral position that is positioned between gear **1** and gear **2**.

(139) Operation of the Shift Mechanism

(140) The following discussion of operation of the shift mechanism **32**, **32′**, **32″** applies to all shift mechanism configurations previously discussed.

(141) Referring to FIG. **4**, the actuator **312** may move the shift collar **310** along the axis **70** between a plurality of positions to selectively couple the shift collar **310** to the transmission **204** or to decouple the shift collar **310** from the transmission **204**. For instance, the actuator **312** may move the shift collar **310** along the axis **70** between the first, second, and third positions. The actuator **312** may also move the shift collar **310** along the axis **70** to first and second neutral positions. In the discussion below, reference to connecting or disconnecting a member of the set of drive pinion gears **210** to/from the drive pinion **30** includes direct and indirect connections to and disconnections from the drive pinion **30**. For instance, a member of the set of drive pinion gears **210** may be directly coupled to the drive pinion **30** or indirectly connected to the drive pinion **30** such as via the drive pinion extension **90**.

(142) In the first position, the shift collar **310** may couple the fourth gear **226** to the drive pinion **30**. For example, the teeth of the shift collar gear **344** may mesh with the inner gear teeth **236** of the fourth gear **226**. Torque may be transmitted from the rotor **106** to the first gear **220** such as via the rotor output flange **130**, from the first gear **220** to the first countershaft gears **270**, **270′**, from the first countershaft gears **270**, **270′** to the fourth countershaft gears **276**, **276′** via the first and second countershafts **260**, **260′**, respectively, from the fourth countershaft gears **276**, **276′** to the fourth gear **226**, and from the fourth gear **226** to the drive pinion **30** via the shift collar gear **344** of the shift collar **310**. The first gear **220**, the second gear **222**, and the third gear **224** may be rotatable about

the axis **70**. Torque may be provided at the first gear ratio in the first position, such as a low-speed gear ratio.

(143) In the first neutral position, the shift collar **310** may not couple any member of the set of drive pinion gears **210** to the drive pinion **30**. As such, the teeth of the shift collar gear **344** may be spaced apart from the first gear **220**, the second gear **222**, the third gear **224**, and the fourth gear **226**. The teeth of the shift collar gear **344** may be axially positioned between the inner gear teeth **234** of the third gear **224** and the inner gear teeth **236** of the fourth gear **226**. As such, the first gear **220**, the second gear **222**, the third gear **224**, and the fourth gear **226** may be rotatable about the axis **70** with respect to the drive pinion **30** when the shift collar **310** is in the first neutral position and torque may not be transmitted between the transmission **204** and the drive pinion **30**. The first neutral position may be positioned between the first position and the second position.

(144) In the second position, the shift collar **310** may couple the third gear **224** to the drive pinion **30**. For example, the teeth of the shift collar gear **344** may mesh with the inner gear teeth **234** of the third gear **224**. Torque may be transmitted from the rotor **106** to the first gear **220** such as via the rotor output flange **130**, from the first gear **220** to the first countershaft gears **270**, **270′**, from the first countershaft gears **270**, **270′** to the third countershaft gears **274**, **274′** via the first and second countershafts **260**, **260′**, respectively, from the third countershaft gears **274**, **274′** to the third gear **224**, and from the third gear **224** to the drive pinion **30** via the shift collar gear **344** of the shift collar **310**. As such, the first gear **220**, the second gear **222**, and the fourth gear **226** may be rotatable about the axis **70** with respect to the drive pinion **30** when the second gear ratio is provided. Torque may be provided at the second gear ratio in the second position, such as a mid-speed gear ratio.

(145) In the second neutral position, the shift collar **310** may not couple any member of the set of drive pinion gears **210** to the drive pinion **30**. As such, the teeth of the shift collar gear **344** may be spaced apart from the first gear **220**, the second gear **222**, the third gear **224**, and the fourth gear **226**. The teeth of the shift collar gear **344** may be axially positioned between the inner gear teeth **234** of the third gear **224** and the inner gear teeth **232** of the second gear **222**. As such, the first gear **220**, the second gear **222**, the third gear **224**, and the fourth gear **226** may be rotatable about the axis **70** with respect to the drive pinion **30** when the shift collar **310** is in the second neutral position and torque may not be transmitted between the transmission **204** and the drive pinion **30**. The second neutral position may be positioned between the second position and the third position.

(146) In the third position, the shift collar **310** may couple the second gear **222** to the drive pinion **30**. For example, the teeth of the shift collar gear **344** may mesh with the inner gear teeth **232** of the second gear **222**. Torque may be transmitted from the rotor **106** to the first gear **220** such as via the rotor output flange **130**, from the first gear **220** to the first countershaft gears **270**, **270′**, from the first countershaft gears **270**, **270′** to the second countershaft gears **272**, **272′** via the first and second countershafts **260**, **260′**, respectively, from the second countershaft gears **272**, **272′** to the second gear **222**, and from the second gear **222** to the drive pinion **30** via the shift collar gear **344** of the shift collar **310**. The shift collar gear **344** may not engage the inner gear teeth **234** of the third gear **224** or the inner gear teeth **236** of the fourth gear **226**. As such, the first gear **220**, the third gear **224**, and the fourth gear **226** may be rotatable about the axis **70** with respect to the drive pinion **30** when the third gear ratio is provided. Torque may be provided at the third gear ratio in the third position, such as a high-speed gear ratio.

(147) The present invention may provide a shift mechanism that has a drive member that may allow the shift collar to be actuated independent of the actuator. This may allow the shift collar to be manually actuated when the actuator is inoperative, such as in a situation in which electrical power is not available to the actuator, when the actuator is stuck or partially stuck and is impaired from moving along the axis, or when a system fault exists that intentionally prevents the actuator from operating. The present invention provides a drive member that is accessible from outside the axle assembly to manually adjust or actuate the shift collar to a desired position. This may allow

the shift collar to be manually actuated to a neutral position to facilitate towing of the vehicle or to permit the transmission gears to rotate freely. The present invention may also facilitate maintenance of the vehicle and adjustment of the position of the shift collar to facilitate proper functioning of the shift mechanism.

(148) While exemplary embodiments are described above, it is not intended that these embodiments describe all possible forms of the invention. Rather, the words used in the specification are words of description rather than limitation, and it is understood that various changes may be made without departing from the spirit and scope of the invention. Additionally, the features of various implementing embodiments may be combined to form further embodiments of the invention.

Claims

1. An axle assembly comprising: a drive pinion that is rotatable about an axis; a transmission that includes a set of drive pinion gears that is spaced apart from the drive pinion and is rotatable about the axis; and a shift mechanism that includes: a shift collar that is rotatable about the axis with the drive pinion and moveable along the axis with respect to the drive pinion; an actuator that is configured to move the shift collar along the axis to selectively connect a member of the set of drive pinion gears to the drive pinion; a linkage that is operatively connected to the actuator and the shift collar and is rotatable about an actuator axis, the linkage having a linkage gear; and a drive member that is rotatable about a drive member axis, wherein the drive member has a drive member gear that has teeth that mesh with teeth of the linkage gear, wherein the shift collar is moveable along the axis when the actuator rotates the linkage and when the drive member is manually actuated to rotate the linkage.
2. The axle assembly of claim 1 wherein the linkage gear is a sector gear that has teeth that extend away from the actuator axis.
3. The axle assembly of claim 1 wherein the drive member has a drive member engagement feature that is engageable with a tool that is configured to manually actuate the drive member.
4. The axle assembly of claim 3 wherein the drive member gear and the drive member engagement feature are disposed at opposite ends of the drive member.
5. The axle assembly of claim 1 wherein the shift mechanism is received in a shift mechanism housing and the drive member extends through a hole in the shift mechanism housing.
6. The axle assembly of claim 5 wherein the shift mechanism housing defines a bore that extends from the hole, and wherein a cap is receivable in the bore to conceal the drive member.
7. An axle assembly comprising: a drive pinion that is rotatable about an axis; a transmission that includes a set of drive pinion gears that is spaced apart from the drive pinion and is rotatable about the axis; and a shift mechanism that includes: a shift collar that is rotatable about the axis with the drive pinion and moveable along the axis with respect to the drive pinion; an actuator that is configured to move the shift collar along the axis to selectively connect a member of the set of drive pinion gears to the drive pinion; a detent linkage that is coupled to the actuator and is rotatable about an actuator axis; a linkage that is coupled to the detent linkage, operatively connected to the shift collar, and is rotatable about the actuator axis; and a drive member that is rotatable about the actuator axis with the detent linkage, wherein the shift collar is moveable along the axis when the actuator rotates the linkage and when the drive member is actuated to rotate the linkage.
8. The axle assembly of claim 7 wherein the drive member is received inside the detent linkage.
9. The axle assembly of claim 7 wherein the detent linkage has a detent linkage spline and the drive member has a drive member spline that mates with the detent linkage spline.
10. The axle assembly of claim 9 wherein the drive member has a drive member engagement feature that is configured to be engaged by a tool that is configured to manually actuate the drive

member.
11. The axle assembly of claim 10 wherein the drive member spline and the drive member engagement feature are disposed at opposite ends of the drive member.
12. The axle assembly of claim 7 wherein the shift mechanism is received in a shift mechanism housing and the drive member extends through a hole in the shift mechanism housing.
13. The axle assembly of claim 12 wherein the shift mechanism housing defines a bore that extends from the hole, and wherein a cap is receivable in the bore to conceal the drive member.
14. The axle assembly of claim 12 wherein a sensor is coupled to the drive member and generates a signal indicative of a rotational position of the drive member.
15. The axle assembly of claim 14 wherein the sensor is mounted to the shift mechanism housing and is disposed outside the shift mechanism housing.
16. The axle assembly of claim 14 further comprising an adapter that extends from the sensor to the drive member, wherein the adapter is received inside the drive member and the sensor has a shaft that is received inside the adapter.
17. The axle assembly of claim 7 wherein the detent linkage is positioned along the actuator axis closer to the actuator than the linkage is positioned to the actuator.
18. The axle assembly of claim 17 wherein the detent linkage defines a detent linkage hole that receives the linkage.
19. The axle assembly of claim 17 wherein the drive member defines a drive member hole that receives the linkage.
20. The axle assembly of claim 7 wherein the shift mechanism is received in a shift mechanism housing and the drive member has an indicator mark that provides a visual indication of a position of the shift collar from outside the shift mechanism housing.